

UNIVERSITY OF SCIENCE

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**ACADEMIC COURSE REPORT
MAJOR PROJECT PROPOSAL**

**PRODUCT DETECTION AND TRACKING
ON MANUFACTURING CONVEYOR BELTS**



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Chapter 1

Project Team Information

1.1 Members

Full Name	Student ID
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1.2 Project Title

Computer Vision System for Automated Product Detection, Tracking, and Monitoring on Manufacturing Conveyor Belts.

Chapter 2

Project Overview

2.1 Research Motivation

In the era of Industry 4.0, automating production processes is critical for maintaining corporate competitiveness. Monitoring product flow on conveyor belts is a vital stage in this automation. Currently, many factories still rely on manual labor for counting and inspection, leading to fatigue-induced errors, slow processing speeds, and high operational costs.

There is a practical demand for a system capable of 24/7 operation with high precision to:

- Automate the counting of output products.
- Monitor flow to detect early-stage blockages or misaligned items.
- Collect operational data to optimize assembly line efficiency.

2.2 Problem Statement

- **Input:** Real-time video stream from industrial cameras mounted above the conveyor belt, observing products such as bottles, cartons, or mechanical parts.
- **Output:**
 - Bounding boxes identifying the location of each product.
 - Unique IDs assigned to each product via multi-object tracking.
 - Trajectories and status indicators (Normal/Defective).
 - Statistical reports on total product throughput within specific timeframes.

2.3 Scientific and Practical Significance

- **Scientific Significance:** This project evaluates the performance of state-of-the-art Deep Learning architectures (YOLO) combined with data filtering algorithms (Kalman Filter, Re-ID) in specific industrial environments. It explores model optimization to achieve a balance between accuracy and real-time processing speed.
- **Practical Significance:** It minimizes human error, increases assembly line productivity by 20-30%, provides accurate digital data for Manufacturing Execution Systems (MES/ERP), and serves as a foundation for automated Quality Assurance (QA/QC).

2.4 Challenges and Scope

2.4.1 Technical Challenges

- **Conveyor Speed:** High-speed movement can cause motion blur.
- **Environmental Variance:** Unstable factory lighting and industrial dust accumulation on lenses.
- **Product Similarity:** Identical appearance of products makes unique ID assignment difficult during occlusion.
- **Product Density:** Crowded or overlapping items can confuse detection models.

2.4.2 Project Scope

- **Datasets:** Standard benchmarks including MVTec D2S (instance segmentation), MVTec AD (anomaly detection), and custom data collected via Roboflow.
- **Methodology:** Focus on YOLOv8/v11 for detection and ByteTrack or Deep SORT for tracking, deployed on GPU-accelerated hardware.

Chapter 3

Related Research

3.1 Practical Application Studies

Rather than focusing solely on theoretical algorithms, this project examines two real-world application systems with contexts highly relevant to industrial requirements:

3.1.1 Product Quality Inspection and Counting System (MDPI, 2024)

The research "**Computer-Vision-Based Product Quality Inspection and Novel Counting System**" presents a computer vision system deployed on an actual production line.

- **Application:** The system performs dual tasks: counting products and detecting surface defects by combining **YOLOv8 Pose** and **difference image** techniques.
- **Results:** Achieved **99.2% accuracy** for products over 1kg by integrating visual data with weight sensors, operating stably at a camera frame rate of 120 FPS.
- **Reference:** *Applied Sciences, MDPI (2024).*

3.1.2 Zone-Based Product Counting System (ResearchGate, 2026)

The study "**Real-Time Object Counting on Industrial Conveyor Belts Using YOLOv11 and Zone-Based Algorithm**" proposes a solution for conveyor belts with high object density.

- **Application:** Utilizes the **YOLOv11** model combined with a **Three-zone counting** algorithm. Instead of traditional line-crossing logic, the system tracks object transitions through three defined spatial zones.
- **Results:** Effectively resolved the "double-counting" phenomenon and achieved **94.2% accuracy** even under 50-70% occlusion.
- **Reference:** *ResearchGate (2026).*

3.1.3 Comparative Analysis of Applied Research

Research Work	Pros	Cons
MDPI System (2024)	Extremely high accuracy (99.2%); Multi-tasking (counting + defects); Stability through weight sensor fusion.	Requires complex hardware (120 FPS cameras, load cells); Sensitive to sudden lighting changes due to difference imaging.
ResearchGate (2026)	Handles high occlusion (50-70%); Eliminates double-counting; YOLOv11 optimized for inference speed.	Requires precise spatial calibration; Performance depends on the stability of tracking within the central zone.

Chapter 4

Proposed Approach

4.1 Identified Problems Post-Survey

Based on the practical research surveyed above, the team has identified two core problems to resolve for system optimization:

1. **Double-Counting and Omissions:** Traditional line-crossing algorithms are prone to errors when objects move inconsistently or when "flickering" occurs between frames due to environmental lighting.
2. **System Latency and ID Switches:** Running complex Deep Learning models on high-speed lines causes inference latency, leading to deviations in the Kalman filter's predicted positions and subsequent ID swaps.

4.2 Proposed Solutions and Improvements

To overcome the aforementioned drawbacks, the following specific improvements are proposed:

- **Tracking Optimization (Inspired by YOLOv11):** Implement a **Virtual Counting Zones** logic. Instead of a single tripwire, the system will establish *Entry*, *Tracking*, and *Exit* zones. A product is only logged into the database once it has successfully navigated all three zones in the correct temporal sequence.
- **Speed Optimization (Inspired by MDPI):** Utilize the **TensorRT** toolkit to optimize YOLO networks, allowing the system to maintain processing speeds above 60 FPS on Edge AI devices (e.g., Jetson Nano), ensuring trajectory continuity.
- **Multi-Task Defect Inspection:** Leverage data from the tracking phase to extract product images at the optimal viewpoint, then apply a lightweight classifier to detect shape abnormalities (dents, deformities) or label defects.

References

1. Jocher, G., et al. (2023). *YOLOv8 by Ultralytics*. <https://github.com/ultralytics/ultralytics>.
2. Zhang, Y., et al. (2022). *ByteTrack: Multi-Object Tracking by Associating Every Detection Box*. In ECCV.
3. Wojke, N., et al. (2017). *Simple Online and Realtime Tracking with a Deep Association Metric*. In ICIP.
4. MVTec Software GmbH. *D2S Dataset for Instance Segmentation*. <https://www.mvtec.com/company/research/datasets/d2s/>.
5. Bergmann, P., et al. (2019). *MVTec AD Dataset for Anomaly Detection*. In CVPR.